Which Adaptation Strategy for a New Task? A Controlled Study of Parameter-Efficient Continual Learning on 3D Brain MRI

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***Abstract*— A department that adds a new analysis task to an existing imaging workflow typically has a few dozen annotated cases, no access to the data behind the tasks already deployed, and a model it cannot afford to retrain or to store many copies of. Which adaptation strategy should it choose? We answer this empirically for 3D brain MRI, comparing eleven strategies—low-rank adaptation (LoRA), bottleneck adapters, bias-only tuning, linear probing, full fine-tuning, and the standard continual-learning remedies EWC, LwF and experience replay—across nine tasks from six public cohorts at 32 labeled volumes each: glioma, stroke-lesion, hippocampus and white-matter-hyperintensity segmentation (BraTS 2023, ISLES 2022, MSD, WMH 2017), brain-age regression and sex classification on two cohorts (IXI, OASIS-1), and dementia classification (OASIS-1), under a subject-grouped held-out protocol with whole-volume inference and five random seeds. The choice turns on one property of the new task: whether it inherits a large task-specific head. All four segmentation tasks do, and there training that decoder directly is at least as accurate as LoRA (by 0.011 Dice on two tasks, equal on two)—at 8.45 M trainable parameters and 32 MB of per-task storage against LoRA's 93.8 K and 366 KB. The global tasks do not, and there only the strategies that move the encoder representation work: LoRA reaches brain-age MAE 10.99 ± 1.27 yr on IXI and 11.95 ± 0.68 yr on OASIS against constant baselines of 14.7 and 23.0, and sex AUC 0.878 ± 0.079 and 0.920 ± 0.035, and bottleneck adapters match or better it, while linear probing and bias-only tuning return a constant output on every one of these tasks despite their frozen features ranking subjects well above chance. We also report three findings that bear directly on practice: freezing shared weights is the only reliable route to zero forgetting, since a second segmentation task erases the first under full fine-tuning in either order (Dice $0.78 \rightarrow 0.25$ and $0.40 \rightarrow 0.01$) and EWC, LwF and replay do not prevent it; a pretrained checkpoint should be validated rather than assumed to transfer, since ours is indistinguishable from random initialisation on glioma segmentation and IXI brain age and helps by at most 0.03 Dice elsewhere; and an evaluation protocol that scores random crops on the model-selection split inflates results enough to reverse conclusions. Full-data references on the four tasks that have more data (908 BraTS volumes; 249 IXI and 106 OASIS subjects) place the 32-shot results 0.09 Dice and 2.7–3.7 yr short of what the same strategies reach with everything, with the ordering between strategies unchanged. Code, splits, per-case predictions and the protocol are released.**

***Index Terms*— Clinical model deployment Few-shot learning Continual learning Parameter-efficient fine-tuning LoRA 3D brain MRI.**

I. Introduction

A hospital imaging service rarely deploys one model and stops. A department running automated tumour segmentation may later want stroke-lesion delineation, then a quantitative marker such as brain age; each capability arrives with a small batch of newly annotated cases, months apart, and typically without access to the data behind the capabilities already in service—retained either not at all, or under governance that makes reuse impractical [1], [2]. The engineering question this poses is concrete: given a frozen model that already works, 32 annotated volumes, and no prior-task data, which adaptation strategy should be used, and what does each cost to run and to store?

The options span four orders of magnitude of trainable parameters. At one end, full fine-tuning updates everything and is the usual default; it also overwrites the representation the earlier tasks depend on, causing *catastrophic forgetting* [3], [4], and requires a separate copy of the whole network per task. Regularisation- and distillation-based remedies such as EWC [3] and LwF [5] aim to soften that at the cost of tuning effort. At the other end, the backbone can be *frozen* and only small task-specific modules trained: a linear probe over frozen features, bias-only tuning [6], or low-rank adaptation (LoRA) [7], [8], which injects trainable low-rank matrices while leaving the original weights untouched. Freezing has an operational attraction beyond compute: because prior-task parameters are never modified, prior-task behaviour cannot change, and because nothing is replayed, no earlier patient data need be retained—a property that matters where retention is governed by HIPAA or GDPR.

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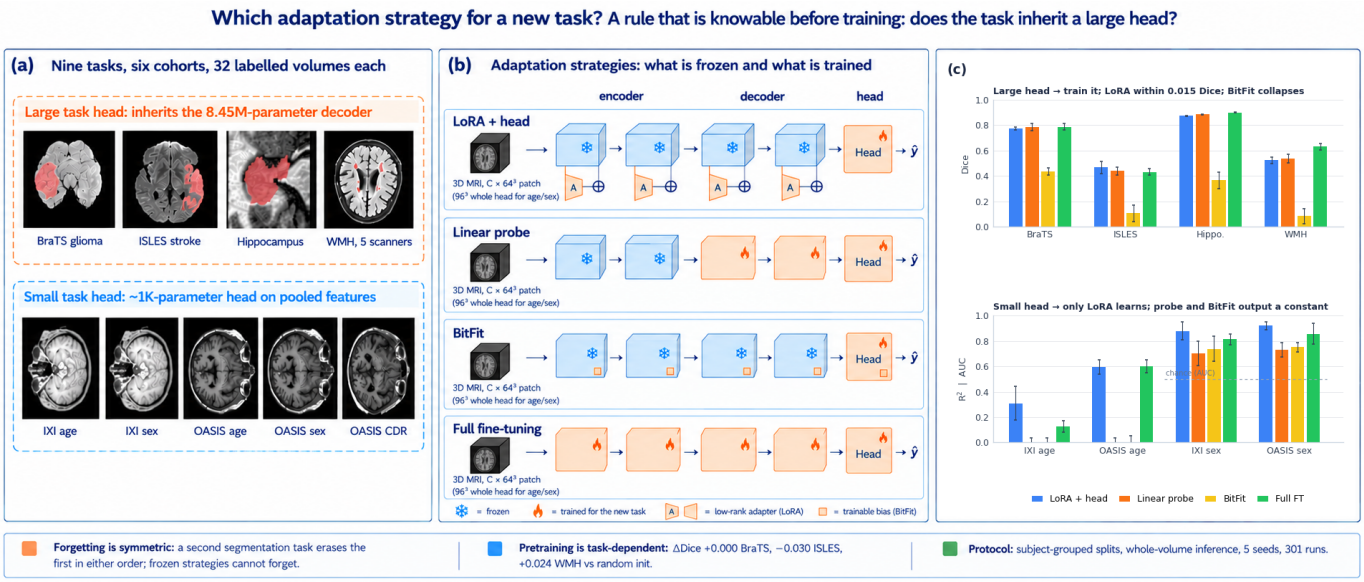


Fig. 1. Overview. (a) Nine brain-MRI tasks from six cohorts, each learned from 32 labelled volumes, grouped by whether the task inherits a large task-specific head (the U-Net decoder) or only a ∼1K-parameter head over pooled encoder features; one held-out test slice per task, segmentation targets in red. (b) The four adaptation strategies compared, drawn as what each freezes (snowflake) and trains (flame) in the same frozen FOMO 3D U-Net: LoRA + head adds low-rank branches to every convolution and trains the output projection; linear probing trains everything downstream of the encoder; BitFit trains bias terms and the head; full fine-tuning trains all 22.6 M parameters. The 66-parameter output projection is trained in every strategy; leaving it frozen is the single largest source of error we observed (Sec. V-B). (c) The decision rule the results support, $N_k = 32$, mean ± SD over five seeds: with a large head, training it is at least as accurate as LoRA and BitFit collapses; with a small head, LoRA is the only frozen strategy that learns at all, while linear probing and BitFit return a constant output. Footer: the three further findings of the paper. Per-case paired tests for every contrast are in Tables III and IV.

Which of these to choose is usually settled by habit rather than evidence, and the evidence that exists is fragmented across single tasks and inconsistent protocols. We therefore ran the comparison directly: eleven strategies, nine tasks spanning segmentation, regression and classification, six public cohorts and five scanners, five seeds, one protocol—357 training runs in all.

The answer turns out to be neither "adapters are efficient" nor "fine-tuning is better", but conditional on a property of the new task that is knowable in advance—whether it inherits a large task-specific head—and the two cases point in opposite directions.

Reaching that answer required first settling how it is measured. The evaluation pipeline we initially built, following what is common in this literature, reported on the same split used for model selection, scored a single random 64^3 crop per case, pooled voxels across cases into one Dice, and let longitudinal timepoints of one patient span training and test. Each of these inflates results, and together they were enough to reverse conclusions: under that pipeline LoRA appeared to underperform linear probing on segmentation by 0.19 Dice, where the true gap is 0.012. We replaced it with subject-grouped train/validation/test splits, whole-volume sliding-window inference, and per-case Dice and 95th-percentile Hausdorff distance (HD95), and report every number below under that protocol. Section IV-B specifies it; we release it with the code because the same failure modes are easy to reproduce accidentally.

Our findings and contributions are:

- **Where a task has no large head, LoRA is necessary, not merely economical—shown on two cohorts.** On whole-head input, brain age reaches MAE 10.99 ± 1.27 yr on IXI and 11.95 ± 0.68 yr on OASIS-1 against constant baselines of 14.7 and 23.0, and sex classification AUC 0.878 and 0.920, with 56K trainable parameters. Frozen-encoder linear probing and BitFit collapse to a constant output on all four tasks, despite the frozen features carrying usable signal (linear-probe sex AUC 0.70–0.73, well above chance): the information is present but unusable without moving the representation. Full fine-tuning of every parameter is worse on IXI (by 1.68 yr and 0.06 accuracy) and matches LoRA on OASIS. Houlsby-style bottleneck adapters, the other way of moving a frozen encoder, learn the same four tasks (OASIS brain age 9.68 yr, sex AUC 0.90–0.92), so the finding is about adapting the representation, not about LoRA's particular form.
- **Where a large task head already exists, LoRA gives up at most a small margin for 90× fewer parameters—shown on four tasks.** Across glioma, stroke-lesion, hippocampus and white-matter-hyperintensity segmentation, full-decoder linear probing leads by 0.011 Dice on BraTS and hippocampus and is statistically indistinguishable on ISLES and WMH, while training 8.45M parameters against 93.8K; at 64 shots the two are equal. Full fine-tuning is the most accurate single-task option on the two tasks without a continual constraint (hippocampus +0.025, WMH +0.108 over LoRA). Bottleneck adapters at the same budget are not a substitute here: they match LoRA on hippocampus and WMH but lose 0.017 Dice

on BraTS and 0.104 on stroke lesions.

- **A 66-parameter configuration detail dominates.** Leaving the segmentation output projection frozen, as "frozen backbone" naively implies, costs 0.248 Dice ($p < 10^{-200}$) and multiplies seed variance tenfold.
- **Forgetting-free is shared, and the standard remedies do not deliver it.** Adapter isolation and frozen-encoder probing both have BWT exactly zero over three-task sequences in which full fine-tuning loses 0.58 Dice in one order and 0.39 in the other—a second segmentation task erases the first whichever comes first. EWC is no better than unregularised fine-tuning (-0.32 vs. -0.33) and LwF and replay are worse (-0.48, -0.40).
- **Pretraining is a small and task-dependent effect.** Against a randomly initialised backbone, the FOMO checkpoint makes no difference on glioma segmentation ($+0.000$ Dice) or IXI brain age (-0.00 yr), is slightly *worse* on stroke lesions (-0.030), and helps by 0.002–0.024 Dice, 1.2 yr and 0.03–0.11 accuracy on the remaining tasks. We therefore claim parameter-efficient adaptation; transfer from this checkpoint is real on some tasks and absent on others.
- **A corrected evaluation protocol and code**, with tests that guard the label-handling and split-leakage failures we encountered—including one that silently zeroed every mask of a single-channel dataset while validation reported a perfect score.

II. Related Work

A. Foundation Models for Medical Imaging

Large-scale pretrained models have transformed medical image analysis. The FOMO initiative [1] provides 3D foundation models pretrained on diverse medical imaging datasets via masked autoencoding objectives, yielding general-purpose representations for downstream adaptation. UNETR [9] introduced Vision Transformers into 3D medical segmentation by replacing CNN encoders with self-attention, demonstrating strong performance on multi-organ and brain tumor segmentation. More recently, SAM-Med3D [10] extended the Segment Anything paradigm to volumetric data, and domain-specific foundation models have emerged for pathology [11] and radiology. These models share a common deployment challenge: adapting to multiple downstream tasks efficiently while preserving the pretrained representations, particularly when labeled data is scarce.

B. Continual Learning in Medical Imaging

Continual learning addresses the problem of sequentially learning new tasks without forgetting previously acquired knowledge [4]. Classical approaches fall into three categories: (i) *regularization-based* methods such as EWC [3] and Synaptic Intelligence [12], which penalize changes to important parameters; (ii) *replay-based* methods such as iCaRL [13], which store or generate exemplars from prior tasks; and (iii) *distillation-based* methods such as LwF [5], which use the previous model's outputs as soft targets. In medical imaging, continual learning has been explored primarily for class-incremental segmentation: LifeLonger [14] benchmarks continual disease classification across dermatology datasets, while Lifelong nnU-Net [15] standardizes continual organ segmentation. However, most prior work considers *class-incremental* settings with homogeneous outputs (e.g., adding new organ classes). Our work addresses *task-incremental* continual learning across *heterogeneous* output types—segmentation, regression and classification—which presents distinct challenges since the task heads and loss functions differ fundamentally, and, as Section V-E shows, the right adaptation strategy differs with them.

C. Parameter-Efficient Fine-Tuning

Parameter-efficient fine-tuning (PEFT) methods adapt large pretrained models by training only a small subset of parameters. LoRA [7] injects trainable low-rank decompositions into weight matrices, achieving competitive performance with full fine-tuning at a fraction of the parameter cost. Alternative PEFT approaches include Adapters [16], which insert bottleneck modules between layers; Prefix-Tuning [17], which prepends learnable tokens to the input; and BitFit [6], which tunes only bias terms. In the context of continual learning, PEFT methods offer a natural mechanism for task isolation: PackNet [18] iteratively prunes and freezes network capacity per task, while Piggyback [19] learns binary masks over a shared backbone. Our approach is most closely related to adapter-isolation strategies but applies LoRA's low-rank structure to 3D convolutions, which is directly compatible with pretrained U-Net architectures. A recurring detail in the PEFT literature, and the one that turns out to matter most here, is that the task-specific output layer is always trained alongside the low-rank updates; a pretrained projection cannot be expected to map onto a new label space.

III. Method

A. Problem Formulation

We consider continual learning with sequential task arrival. Let f_θ denote a pretrained 3D U-Net backbone (frozen) and $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_K$ a sequence of tasks. For each task k, we have access to only N_k labeled few-shot samples ($N_k = 32$ throughout) drawn from task-specific distribution $\mathcal{D}_k = \{(x_i^{(k)}, y_i^{(k)})\}_{i=1}^{N_k}$. We assume *no replay*: when training on $\mathcal{T}_k$, we cannot access data from $\mathcal{T}_{<k}$. The optimization objective for task k is:

$$\min_{\phi_k, h_k} \frac{1}{N_k} \sum_{i=1}^{N_k} \mathcal{L}_k\big(h_k(f_\theta(x_i^{(k)}; \phi_k)),\ y_i^{(k)}\big) \tag{1}$$

where ϕ_k denotes the LoRA adapter parameters for task k, h_k the task-specific head, and $\mathcal{L}_k$ the task loss. Crucially, θ remains frozen throughout; only ϕ_k and h_k are optimized. The goal is to learn $\{\phi_k, h_k\}_{k=1}^K$ such that performance on all seen tasks remains high after training on $\mathcal{T}_K$.

Tasks are of three output types—a 3D binary mask (segmentation), a scalar (regression) or a class (classification)—each with its own modality set (one to three input channels) and

its own output head h_k. Because the modality counts differ, the first convolution of the encoder cannot be shared and is task-specific; all remaining encoder weights (39 of 40 tensors) are shared and frozen.

Figure 1 gives the overview: the nine tasks grouped by head size (a), the four strategies drawn as what each freezes and trains in the shared backbone (b), and the decision rule the results support (c). Under adapter isolation, inference for task k uses backbone f_θ + adapter ϕ_k + head h_k; prior adapters and the backbone remain frozen. Task identity is assumed known at inference.

B. LoRA for 3D Convolutions

Low-Rank Adaptation (LoRA) [7] injects trainable low-rank matrices into a frozen weight matrix $W \in \mathbb{R}^{d\times k}$:

$$W' = W + \Delta W, \quad \Delta W = \frac{\alpha}{r} B \cdot A, \quad A \in \mathbb{R}^{r\times k},\ B \in \mathbb{R}^{d\times r} \tag{2}$$

with rank $r \ll \min(d, k)$ and scaling factor α. Only A and B are trained; W stays frozen. The scaling ratio α/r controls the magnitude of the low-rank perturbation relative to the pretrained weights.

For 3D convolutions in a U-Net, we apply LoRA via parallel $1 \times 1 \times 1$ convolutions acting on the channel dimension:

$$\text{Conv3d}'(x) = \text{Conv3d}(x; W) + \frac{\alpha}{r}\,\text{Conv1}(x; B) \circ \text{Conv1}(x; A) \tag{3}$$

where Conv1 denotes $1 \times 1 \times 1$ convolution, $A \in \mathbb{R}^{r\times C_{\text{in}}}$ projects to rank-r space, and $B \in \mathbb{R}^{C_{\text{out}}\times r}$ projects back. This formulation is memory-efficient and compatible with standard Conv3d layers [20].

We use $\alpha = 16$, evaluate $r \in \{4, 8, 16\}$ (Sec. V-C) and report $r = 16$, and target every Conv3d in both encoder and decoder for segmentation; encoder-only suffices for regression. For a single Conv3d layer with input channels C_{in} and output channels C_{out}, the number of LoRA parameters is:

$$|\phi_{\text{layer}}| = r \cdot (C_{\text{in}} + C_{\text{out}}) \tag{4}$$

which is substantially smaller than the original $C_{\text{out}} \times C_{\text{in}} \times D \times H \times W$ parameters. Summed over all targeted layers at $r = 16$, the adapter for segmentation has 93,810 parameters against a 22.63M-parameter U-Net (0.41%), and the adapter for regression has 55,841 parameters against the 14.16M-parameter encoder-plus-head (0.39%); at $r = 8$ the counts are 46,872 and 28,177.

C. Continual Learning via Adapter Isolation

Each task k receives a *dedicated LoRA adapter* ϕ_k. The full training and inference procedure is summarized in Algorithm 1. The trainable parameter set for task k is strictly disjoint from all other tasks:

$$\Phi_k = \{\phi_k, h_k\}, \quad \Phi_i \cap \Phi_j = \emptyset \ \ \forall\, i \neq j \tag{5}$$

Because θ and $\{\Phi_{<k}\}$ are frozen when optimizing Φ_k, the performance of any prior task $j < k$ is invariant to training on

Algorithm 1 Continual Learning via LoRA Adapter Isolation

Require: Pretrained backbone f_θ; task sequence $\mathcal{T}_1, \ldots, \mathcal{T}_K$; subject-grouped train/val/test splits per task; LoRA rank r; epochs E
Ensure: Trained adapters $\{\phi_k\}_{k=1}^K$ and heads $\{h_k\}_{k=1}^K$

```
Freeze all parameters of f_θ
for k = 1 to K do
    Initialize LoRA adapter φ_k with rank r (Kaiming A, zero B)
    Initialize task head h_k (output projection for segmentation; pooled regressor for regression); mark trainable
    if k > 1 then
        Freeze all previous adapters φ_1, ..., φ_{k-1} and heads h_1, ..., h_{k-1}
    end if
    for epoch = 1 to E do
        for each mini-batch (x, y) ∈ D_k^train do
            ŷ ← h_k(f_θ(x; φ_k))
            Update φ_k, h_k on L_k(ŷ, y) with AdamW
        end for
        Score D_k^val (deterministic centre crop); keep the best epoch
    end for
    Score the kept checkpoint once on D_k^test with whole-volume inference
end for
```

$\mathcal{T}_k$. Formally, let $R_{t,j}$ denote the performance of task j after training up to task t. Then:

$$R_{t,j} = R_{j,j} \quad \forall\, t \geq j \implies \text{BWT}_j = R_{K,j} - R_{j,j} = 0 \tag{6}$$

This zero-forgetting property holds *by construction*, without regularization or replay. It is not, however, unique to adapters: any scheme that leaves shared weights untouched—including linear probing on a frozen encoder—satisfies Eq. (6) equally. What distinguishes the schemes is how many parameters they must train to reach a given accuracy, which is what Section V measures. The total model storage after K tasks is:

$$\text{Storage} = |\theta| + \sum_{k=1}^{K} (|\phi_k| + |h_k|) \tag{7}$$

which grows linearly in K with a small per-task overhead since $|\phi_k| \ll |\theta|$.

D. Task Heads and Losses

Task 1 (Segmentation). The full U-Net decoder produces a voxel-wise two-class prediction. LoRA is applied to every Conv3d in encoder and decoder. The head h_1 is the decoder's final $1\times1\times1$ projection from 32 channels to the 2 output classes (66 parameters), which is trained from its pretrained initialization. We stress this because a natural but wrong reading of "frozen backbone" is to freeze this layer too; Section V-B quantifies the consequence. The training loss is

cross-entropy plus soft Dice on the foreground probability:

$$\mathcal{L}_{\text{seg}} = \text{CE}(p, g) + 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon} \tag{8}$$

where p_i is the softmax foreground probability at voxel i, g_i the binary label, and $\epsilon = 10^{-8}$.

Task 2 (Regression). For brain age estimation we use the encoder only: its deepest feature map is global-average-pooled and passed through a two-layer fully connected head with a single scalar output. LoRA targets the encoder's Conv3d layers. The loss is mean squared error in years.

Optimization. All methods use AdamW (learning rate 10^{-3}, weight decay 10^{-5}) with cosine annealing to 10^{-6} over 100 epochs, batch size 2, gradient-norm clipping at 1.0, and LoRA dropout 0.1. Training patches are random 64^3 crops; no other augmentation is used. Hyperparameters are identical across methods and were not tuned per method.

IV. Experiments

A. Datasets

Notation. We use nine tasks from six cohorts: T1 glioma segmentation, T2 brain-age regression and T3 stroke-lesion segmentation form the continual sequences (Sec. V-G); T4 hippocampus segmentation, T5 sex classification and T6 white-matter-hyperintensity segmentation are single-task evaluations that test whether the two patterns we find replicate on segmentation and global tasks drawn from outside the original pair; and T7 brain age, T8 sex and T9 dementia classification repeat the global tasks on a second cohort (OASIS-1) acquired on a different scanner.

T1: BraTS 2023 Glioma [21]. We use the 1251 labeled volumes of the training release, with channels T2w, T2-FLAIR and T1c and a binary whole-tumor mask derived from the expert annotation. The challenge's validation release ships without segmentations and is excluded; an earlier pipeline had written those 219 volumes with all-zero masks, which are indistinguishable from a tumor-free label and had contaminated 17% of the test set. Several subjects contribute more than one longitudinal timepoint; these are kept in the same split.

T2: IXI [22]. Of the 577 subjects with both T1 and T2 scans, 559 have a usable age in the IXI demographics table (15 have no table entry, 3 have a missing age) and are used; none is imputed. Test-set ages span 20.0–86.3 yr (mean 50.0, SD 16.5).

T3: ISLES 2022 [23]. Ischemic-stroke lesion segmentation with channels DWI, ADC and FLAIR (FLAIR resampled onto the DWI grid) and the expert lesion mask; 247 of the 250 cases are used, three being excluded because their mask emptied under 1 mm resampling. Lesions are small (median $\approx$775 voxels), which makes this a harder segmentation task than T1 and one drawn from a single cohort, so unlike an earlier infarct-*classification* design that contrasted ISLES with IXI controls, it carries no acquisition confound.

T2 (whole-head variant). Because patch input proved to be the limiting factor for regression (Sec. V-D), we also prepare each IXI subject as a single 96^3 cube: the volume is cropped to its non-zero content and resampled, so one forward pass sees the entire head. The task, labels, splits and metrics are otherwise identical, and models are run with a 96^3 window, which makes the crop the identity and evaluation a single window.

T4: MSD Task04 hippocampus [24]. 260 single-channel T1 volumes with anterior and posterior hippocampus labels, merged to binary. This task shares none of the incidental properties of T1 and T3: one input channel rather than three, healthy anatomy rather than pathology, a large consistent foreground (4.7% of voxels against $<$0.1% for a lesion), and volumes of roughly $35\times51\times35$ that fit inside the 64^3 window, so the whole structure is visible in one forward pass. It therefore tests whether the segmentation result is about the presence of a large decoder or about tumours.

T5: IXI sex classification. The same 559 subjects, scans, modalities and whole-head 96^3 cubes as T2, labelled instead by the sex recorded in the IXI demographics table (247 male, 312 female; majority-class accuracy 0.56). It differs from T2 only in output type, which makes it a direct test of whether the regression finding is about the output or about where capacity must sit. It contributes the study's classification task; an infarct-detection formulation contrasting two cohorts was rejected as confounded by acquisition (Sec. VI-B).

T6: WMH Segmentation Challenge 2017 [25]. 170 subjects from three sites and five scanners (Utrecht 3T Philips; Singapore 3T Siemens; Amsterdam 3T GE, 1.5T GE and 3T Philips PET-MR), each with a FLAIR and a co-registered T1 and an expert white-matter-hyperintensity mask; the challenge's "other pathology" label is mapped to background, as in its own scoring. It is the only task whose test set spans several scanners, so it adds a multi-site acquisition shift to the segmentation family as well as a fourth pathology.

T7–T9: OASIS-1 [26]. 416 subjects aged 18–96 with a single 1.5T Siemens T1 (the challenge-supplied N4-corrected, 1 mm native-space average of the MPRAGE repeats), prepared as 96^3 whole-head cubes exactly as the IXI tasks are. T7 repeats brain-age regression and T8 sex classification (160 male, 256 female) on this second cohort; T9 labels the 198 subjects aged 60 and over by Clinical Dementia Rating, 100 with CDR$>$0 against 98 at CDR 0, a clinical classification that IXI cannot provide. The 20 reliability re-scans are dropped so every subject appears once.

Data use. BraTS 2023 was obtained from Synapse (syn51156910) under its Data Use Agreement; IXI is publicly available at `https://brain-development.org/ixi-dataset/`; ISLES 2022 is distributed under CC BY 4.0 (Zenodo 7153326); WMH 2017 under CC BY-NC 4.0 (DOI 10.34894/AECRSD); OASIS-1 under its Data Use Agreement (`https://sites.wustl.edu/oasisbrains/`). No additional ethics approval was required.

Preprocessing. Volumes are resampled to 1 mm isotropic and z-scored per channel over non-zero voxels.

B. Splits and Evaluation Protocol

For each task and seed we draw a subject-grouped three-way split: a training set of exactly $N_k = 32$ volumes, a validation

set of 60 subjects used only for epoch selection, and a held-out test set of up to 250 subjects (275 volumes for T1; 250 for T2, T5, T7 and T8; all remaining subjects elsewhere: 155 for T3, 168 for T4, 78 for T6, 106 for T9) on which every number in this paper is reported. No subject appears in more than one split. Splits are persisted so that all methods are scored on identical cases, which is what makes the paired comparisons below valid.

Segmentation is scored on the whole volume by Gaussian-weighted sliding-window inference (64^3 windows, 25% overlap), followed by argmax. We report per-case Dice and per-case HD95 in voxel units, averaged over cases. **Regression** is scored by averaging the prediction over a fixed $2\times2\times2$ grid of 64^3 windows. We report MAE in years, R^2, and the standard deviation of the predictions, the last because it exposes a model that has collapsed to a constant regardless of MAE.

Backward transfer. For the sequential baselines, after each later task we re-score every earlier task's test set with that task's own head on the now-updated encoder and report $\mathrm{BWT}_j = R_{K,j} - R_{j,j}$ in Dice; in the three-task sequence we also report the intermediate value after T3. For adapter-isolated methods this quantity is zero by construction and is not re-measured. Both sequences are also run in reverse order (T2→T1 and T2→T3→T1) to test whether forgetting depends on which task comes first.

Statistics. Means and SDs are over seeds. Paired comparisons pool per-case differences across seeds (all methods share the split within a seed) and report the mean difference with a bootstrap 95% CI (10,000 resamples) and a Wilcoxon signed-rank p; classification accuracy is compared with McNemar's exact test. Within each table of contrasts, p-values are Holm-adjusted for the number of contrasts in that table and both raw and adjusted values are shown.

C. Backbone

`unet_b` from FOMO [1], initialized from `fomo25_mmunetvae_pretrained.ckpt`. All 40 encoder tensors except the modality-specific input convolution load from the checkpoint; the decoder loads 59 of 82 tensors, the remainder being normalization parameters. The same checkpoint is loaded at the start of every task for every method.

D. Compared Methods

All methods share the backbone, data, splits, optimizer and schedule; they differ only in which parameters are trainable (Table I).

- **LoRA (enc+dec).** Rank-8 LoRA on every Conv3d with the output projection left frozen—the configuration a literal reading of "freeze the backbone" produces. Included to quantify that reading (Sec. V-B).
- **LoRA + head (proposed).** As above, plus the 66-parameter output projection. Reported at rank 16; ranks 4 and 8 appear in the ablation of Sec. V-C.
- **Sequential Linear.** Encoder frozen; every downstream parameter trainable. For segmentation this is the entire decoder (8.45M parameters); for regression, the 513-parameter head. One model carried through the sequence.
- **Sequential FT.** All parameters trainable; one model carried through the sequence. The forgetting reference.
- **BitFit** [6]. PEFT comparator below LoRA's budget: bias terms plus the output projection (6.4K parameters), task-isolated like LoRA.
- **Bottleneck adapters** [16]. PEFT comparator at LoRA's budget. Where LoRA adds a low-rank *parallel* update to a frozen convolution, a Houlsby adapter adds a small *serial* bottleneck after it: $y \leftarrow y + U\,\mathrm{GELU}(D\,y)$ with D a $1\times1\times1$ projection to 16 channels and U back, U zero-initialised so training starts from the frozen network's function. Inserted after every $3\times3\times3$ convolution LoRA targets, plus the output projection: 97.5K parameters for segmentation and 66.1K for the global tasks (LoRA r=16: 93.8K and 56.4K), task-isolated like LoRA.
- **EWC** [3], **LwF** [5], **experience replay**. Full fine-tuning plus a forgetting-mitigation term: a diagonal-Fisher penalty ($\lambda = 10^3$), encoder-feature distillation from the previous task's model (weight 0.5), and the same distillation on a 100-sample buffer of previous-task volumes, respectively.
- **Random-initialisation control.** LoRA + head (r=16) with the backbone initialised randomly instead of from the checkpoint, to test whether pretraining contributes anything (Sec. V-I).
- **Full-data reference.** LoRA + head (r=16), Sequential Linear and Sequential FT trained on every labelled subject outside the seed-42 validation and test sets—908 BraTS volumes (10 epochs, about 2.8× the few-shot gradient budget), 249 IXI and 106 OASIS subjects (the 100-epoch recipe unchanged)—one seed, to show how far the 32-shot results sit below what the same strategies reach with all the data (Sec. V-F). ISLES, hippocampus, WMH and CDR have no subjects outside the few-shot protocol's validation and test sets, so no such reference exists for them.

Per-task oracles (one independent model per task, no continual constraint) were run only under the superseded protocol and are omitted rather than reported on a different footing.

TABLE I
TRAINABLE PARAMETERS PER METHOD. THE COMPARISON IN TABLE II SPANS FOUR ORDERS OF MAGNITUDE OF TRAINABLE BUDGET.

	T1 segmentation		T2 regression	
Method	trainable	share	trainable	share
LoRA (enc+dec), r=8	46,872	0.207%	28,177	0.199%
LoRA + head, r=8	46,938	0.207%	28,177	0.199%
LoRA + head, r=16	93,810	0.414%	55,841	0.393%
Bottleneck adapters	97,506	0.430%	66,145	0.466%
Sequential Linear	8,446,306	37.4%	513	0.004%
Sequential FT	22,582,978	100%	14,136,321	100%

E. Implementation and Resources

Experiments run on a single NVIDIA RTX 3090. Measured peak GPU memory is 1.67 GB during segmentation training

(batch 2) and 4.16 GB during whole-volume evaluation (batch 8 windows); regression peaks at 0.69 GB and 2.53 GB respectively. One LoRA segmentation run trains in 41.5 min (Table XI) and evaluates 275 volumes in about 25 min; a whole-head regression run trains and evaluates in under 10 min. Code, splits and per-case predictions are released with the paper.

V. Results

All results are on held-out test splits at $N_k = 32$, reported as mean $\pm$ SD over five seeds (42–46; the reversed-order sequences use three); each seed draws its own subject-grouped split, and all methods share the split within a seed. Paired comparisons pool per-case differences across seeds ($n = 1366$ for BraTS, 1250 for brain age and sex, 840 for hippocampus, 775 for ISLES, 390 for WMH). All 357 runs completed; none failed.

A. Main Comparison

Table II gives the comparison on the two-task sequence and Table III the paired tests.

B. The Output Projection Accounts for Most of the LoRA Gap

At rank 8, with the decoder's final $1\times1\times1$ projection frozen, LoRA on every other layer reaches Dice 0.503 ± 0.117 and HD95 135 ± 17; the spread across seeds reflects how much the outcome depends on whether the pretrained projection happens to be usable for the new label. Training those 66 additional parameters lifts it to 0.750 ± 0.027 and HD95 48.8 ± 4.6—a paired gain of $+0.248$ Dice ($p = 6\times10^{-222}$) at a budget that changes from 46,872 to 46,938, and a fourfold reduction in seed SD. Mechanically the pretrained projection maps decoder channels onto the pretraining objective's output space, and rank-8 perturbations upstream cannot re-map it onto a new label space. Every PEFT recipe trains the new head; a "frozen backbone" should not be read as including it.

C. Accuracy per Trainable Parameter, and Rank

Sequential Linear remains more accurate than LoRA + head at rank 16, but narrowly: Dice 0.782 ± 0.032 vs. 0.772 ± 0.011, a paired -0.011 (95% CI $[-0.014, -0.007]$, $p = 6\times10^{-6}$), and HD95 43.5 vs. 44.8, with LoRA the more seed-stable of the two. It reaches that accuracy by training the full 8.45M-parameter decoder; LoRA + head trains 93.8K, a factor of 90 fewer, for a 1.4% relative Dice cost. Full fine-tuning is better still, by $+0.015$ over LoRA + head, at $240\times$ the budget and at the cost of forgetting (Sec. V-G).

Table V varies the rank. Rank 4 is not worse than rank 8 (0.754 vs. 0.750) at half the parameters, so rank 8 sits in a flat region; rank 16 gains $+0.022$ over rank 8 ($p = 4\times10^{-79}$) and cuts the seed SD from 0.027 to 0.011. BitFit, at 6.4K parameters, is far too small for this task (0.436) and marks the lower end of the budget curve.

The gap also closes with data (Table VI, Fig. 2): at 16 shots linear probing leads by 0.101, at 32 by 0.011, and at 64 the two are equal (0.816 both); with all 908 volumes they are 0.859 and 0.849 (Sec. V-F). The parameter advantage of LoRA is therefore paid for only in the smallest-data regime—the opposite of the intuition that adapters are the few-shot method.

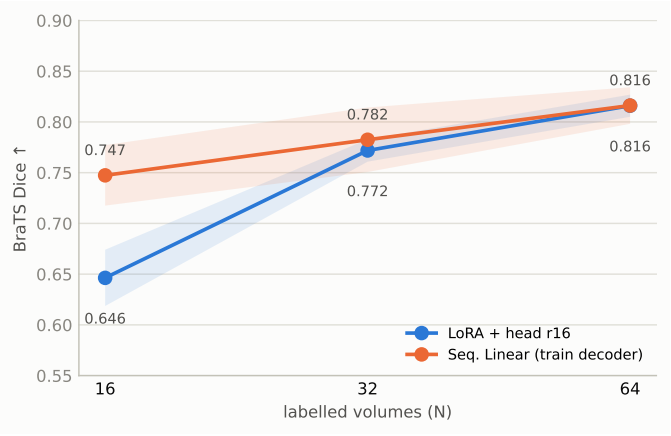


Fig. 2. BraTS Dice against the number of labelled volumes for LoRA + head ($\boldsymbol{r=16}$) and Sequential Linear (decoder trained on a frozen encoder); mean $\pm$ SD over five seeds. Training the decoder leads by 0.10 Dice at 16 shots and the gap is gone at 64.

D. Brain Age: Input Scope Decides, and LoRA Wins

From 64^3 patches, no method learns brain age: all ten configurations in Table II land within 14.3–14.7 yr of the constant predictor (14.4 yr) with $R^2 \approx 0$, including full fine-tuning of every parameter. A 64^3 window covers a few percent of a $240\times240\times180$ head, and the cues for age—ventricular volume, cortical thinning, global atrophy—are not local.

Re-running the task with the whole head resampled to a single 96^3 cube changes the picture entirely (Table VII). LoRA + head reaches MAE 10.99 ± 1.27 yr with $R^2 = 0.31$, beating the constant predictor by 3.68 yr per subject (paired, $p = 3\times10^{-41}$) and full fine-tuning by 1.68 yr ($p = 3\times10^{-13}$; Table IV). Sequential Linear and BitFit do *not* recover: their predictions still have a standard deviation of about 0.1 yr against a true 16.5, i.e. they remain constant.

Figure 3 shows what the numbers summarise: the linear probe's predictions sit in a 47–54 yr band whatever the true age, while LoRA's follow it, on both cohorts.

This is the paper's clearest result and it runs opposite to segmentation. Brain age needs the *encoder* representation to change—a 513-parameter head on frozen features cannot express it, and bias-only adaptation cannot either—while 56K well-placed low-rank parameters can, and do so better than releasing all 14M. Where segmentation already has a large task-specific decoder to train, regression has nothing comparable, and that is exactly where adapter capacity becomes necessary rather than merely economical.

E. Both Patterns Replicate on Held-Out Task Families

The two findings above each rested on a single task and a single cohort. Table VIII and Fig. 5 add two segmentation tasks and four global tasks chosen to share as little as possible with the originals—a different pathology, a different site and

TABLE II

HELD-OUT TEST RESULTS, $N_k = 32$, MEAN ± SD OVER SEEDS 42–46, SEQUENCE T1→T2. T1 = BRATS GLIOMA SEGMENTATION (275 VOLUMES, 250 SUBJECTS PER SEED); T2 = IXI BRAIN AGE FROM 64^3 PATCHES (250 SUBJECTS). "T1 AFTER T2" IS T1 RE-SCORED AFTER THE MODEL HAS LEARNED T2; BWT IS ITS CHANGE. CONSTANT-PREDICTOR MAE FOR T2 IS ≈14.4 YR.

	T1 segmentation		T2 brain age (patch)		Forgetting		
Method	Dice↑	HD95↓	MAE (yr)↓	R^2	T1 after T2	BWT	Trainable
Frozen backbone (task-isolated; BWT = 0 by construction)							
BitFit	0.436 ± 0.031	77.1 ± 9.3	14.66 ± 0.24	-0.03	= T1	0.000	6,434
LoRA (enc+dec), r=8	0.503 ± 0.117	135.1 ± 17.1	14.47 ± 0.35	-0.04	= T1	0.000	46,872
LoRA + head, r=4	0.754 ± 0.032	49.0 ± 6.6	14.32 ± 0.35	-0.00	= T1	0.000	23,502
LoRA + head, r=8	0.750 ± 0.027	48.8 ± 4.6	14.43 ± 0.27	-0.01	= T1	0.000	46,938
LoRA + head, r=16	0.772 ± 0.011	44.8 ± 2.4	14.67 ± 0.49	-0.05	= T1	0.000	93,810
Sequential Linear	0.782 ± 0.032	43.5 ± 7.1	14.71 ± 0.23	-0.04	0.782 ± 0.032	0.000	8,446,306
Backbone updated per task							
Sequential FT	0.787 ± 0.028	43.0 ± 6.5	14.65 ± 0.27	-0.02	0.455 ± 0.149	-0.331 ± 0.148	22,582,978
EWC	0.784 ± 0.027	43.7 ± 5.7	14.65 ± 0.23	-0.03	0.463 ± 0.157	-0.321 ± 0.181	22,582,978
LwF	0.789 ± 0.028	42.4 ± 6.4	14.69 ± 0.22	-0.03	0.315 ± 0.162	-0.475 ± 0.171	22,582,978
Replay	0.754 ± 0.043	50.1 ± 9.2	14.64 ± 0.30	-0.03	0.354 ± 0.141	-0.400 ± 0.103	22,582,978

TABLE III

PAIRED PER-CASE DIFFERENCES IN DICE ON THE FOUR SEGMENTATION TASKS, POOLED OVER FIVE SEEDS (BOOTSTRAP 95% CI; WILCOXON SIGNED-RANK p, RAW AND HOLM-ADJUSTED WITHIN THIS TABLE). L+H$_r$ = LORA + HEAD AT RANK r; SL = SEQUENTIAL LINEAR; FT = SEQUENTIAL FT; L$_8$ = LORA (ENC+DEC) WITH THE OUTPUT PROJECTION FROZEN; RI = THE SAME LORA + HEAD CONFIGURATION ON A RANDOMLY INITIALISED BACKBONE; AD = BOTTLENECK ADAPTERS. ISLES SL AND FT ARE SCORED AT LEARN INSIDE THE THREE-TASK SEQUENCE; THE ISLES RI AND AD CONTRASTS USE SINGLE-TASK RUNS.

Contrast	ΔDice	95% CI	p	p_{Holm}	n
BraTS: L+h$_{16}$ − SL	−0.011	[−0.014, −0.007]	6×10^{-6}	4×10^{-5}	1366
BraTS: FT − L+h$_{16}$	+0.015	[+0.012, +0.018]	8×10^{-33}	9×10^{-32}	1366
BraTS: L+h$_{16}$ − L+h$_8$	+0.022	[+0.019, +0.024]	4×10^{-79}	6×10^{-78}	1366
BraTS: L+h$_8$ − L$_8$	+0.248	[+0.240, +0.257]	6×10^{-222}	1×10^{-220}	1366
BraTS: L+h$_{16}$ − RI	+0.000	[−0.002, +0.002]	0.23	0.69	1366
ISLES: SL − L+h$_{16}$	−0.002	[−0.009, +0.005]	0.01	0.07	775
ISLES: FT − L+h$_{16}$	−0.009	[−0.018, −0.000]	0.86	1.00	775
ISLES: L+h$_{16}$ − RI	−0.030	[−0.038, −0.022]	1×10^{-12}	1×10^{-11}	775
Hippocampus: SL − L+h$_{16}$	+0.011	[+0.010, +0.012]	2×10^{-70}	3×10^{-69}	840
Hippocampus: FT − L+h$_{16}$	+0.025	[+0.024, +0.027]	3×10^{-134}	6×10^{-133}	840
Hippocampus: L+h$_{16}$ − RI	+0.002	[+0.001, +0.002]	2×10^{-4}	1×10^{-3}	840
WMH: SL − L+h$_{16}$	+0.015	[+0.004, +0.025]	0.08	0.33	390
WMH: FT − L+h$_{16}$	+0.108	[+0.098, +0.117]	9×10^{-61}	1×10^{-59}	390
WMH: L+h$_{16}$ − RI	+0.024	[+0.017, +0.030]	1×10^{-10}	9×10^{-10}	390
BraTS: L+h$_{16}$ − AD	+0.017	[+0.014, +0.020]	5×10^{-16}	6×10^{-15}	1366
ISLES: L+h$_{16}$ − AD	+0.104	[+0.093, +0.115]	5×10^{-61}	7×10^{-60}	775
Hippocampus: L+h$_{16}$ − AD	−0.002	[−0.003, −0.001]	2×10^{-7}	2×10^{-6}	840
WMH: L+h$_{16}$ − AD	−0.003	[−0.011, +0.005]	0.64	1.00	390

scanner, a second cohort for both global tasks, and a clinical label—and both patterns hold.

Segmentation. On hippocampus segmentation—one channel, healthy anatomy, a foreground two orders of magnitude larger than a lesion, and volumes that fit inside the inference window—linear probing again leads LoRA, by 0.011 Dice (paired, 95% CI $[0.010, 0.012]$, $p = 2\times10^{-70}$). On white-matter hyperintensities, whose test set spans five scanners, the two are statistically indistinguishable ($+0.015$, $p_{\text{Holm}} = 0.33$), as they are on stroke lesions (-0.002, $p_{\text{Holm}} = 0.07$). Across the four segmentation tasks the margin is therefore 0.011, 0.011, 0.000 and 0.015 (Table III): never in LoRA's favour, never larger than 0.015 Dice, across tasks that differ in modality count, pathology, foreground fraction, input scope and acquisition. The property that predicts it is the one they share—an 8.45M-parameter decoder that is already task-shaped and can simply be trained. Full fine-tuning, which has no continual constraint on a single task, is the most accurate option on hippocampus ($+0.025$) and by a wide margin on WMH ($+0.108$), at the cost of a separate 86 MB network per task.

Figure 4 shows representative cases: on each task the two frozen-encoder strategies produce masks of similar quality, differing in detail rather than in kind, which is what a 0.011-Dice margin looks like.

Global tasks. On IXI sex classification, LoRA reaches AUC 0.878 ± 0.079 and per-case accuracy 0.06 above full fine-tuning's (McNemar $p = 2\times10^{-4}$). Linear probing and BitFit again produce a *constant* decision: balanced accuracy 0.500 on every seed. Their AUCs, however, are 0.70 and 0.74—well above chance. This sharpens the mechanism: the frozen features do carry sex information, enough to rank subjects correctly most of the time, but a 1K-parameter head over pooled features cannot turn that ranking into a usable decision boundary. Moving the representation is what converts present-but-unusable signal into a working classifier.

TABLE IV

PAIRED PER-CASE CONTRASTS ON THE GLOBAL TASKS, POOLED OVER FIVE SEEDS ($n = 1250$ CASES; 530 FOR CDR). BRAIN AGE: DIFFERENCE IN ABSOLUTE ERROR IN YEARS (NEGATIVE FAVOURS THE FIRST METHOD; "CONSTANT" PREDICTS THE MEAN TRAINING AGE). SEX AND CDR: DIFFERENCE IN PER-CASE ACCURACY AT A 0.5 THRESHOLD, MCNEMAR'S EXACT TEST. p_{HOLM} IS ADJUSTED WITHIN EACH BLOCK.

Contrast	Δ	95% CI	p	p_{Holm}
Brain age, Δ|error| (yr)				
IXI: L+h$_{16}$ − constant	-3.683	$[-4.172, -3.197]$	3×10^{-41}	2×10^{-40}
IXI: L+h$_{16}$ − FT	-1.683	$[-2.089, -1.284]$	3×10^{-13}	2×10^{-12}
IXI: L+h$_{16}$ − RI	-0.002	$[-0.370, +0.374]$	0.51	0.92
OASIS: L+h$_{16}$ − constant	-10.758	$[-11.573, -9.942]$	6×10^{-97}	5×10^{-96}
OASIS: L+h$_{16}$ − FT	-0.049	$[-0.595, +0.490]$	0.46	0.92
OASIS: L+h$_{16}$ − RI	-1.153	$[-1.708, -0.600]$	1×10^{-7}	6×10^{-7}
IXI: L+h$_{16}$ − AD	-0.490	$[-0.887, -0.102]$	0.04	0.13
OASIS: L+h$_{16}$ − AD	$+2.263$	$[+1.758, +2.770]$	1×10^{-17}	9×10^{-17}
Classification, Δaccuracy				
IXI sex: L+h$_{16}$ vs FT	$+0.062$	$[+0.030, +0.094]$	2×10^{-4}	1×10^{-3}
IXI sex: L+h$_{16}$ vs RI	$+0.109$	$[+0.082, +0.135]$	2×10^{-15}	1×10^{-14}
OASIS sex: L+h$_{16}$ vs FT	-0.031	$[-0.060, -0.002]$	0.04	0.12
OASIS sex: L+h$_{16}$ vs RI	$+0.030$	$[+0.010, +0.050]$	3×10^{-3}	0.01
OASIS CDR: L+h$_{16}$ vs FT	-0.028	$[-0.089, +0.030]$	0.38	0.38
IXI sex: L+h$_{16}$ vs AD	-0.020	$[-0.045, +0.004]$	0.13	0.26
OASIS sex: L+h$_{16}$ vs AD	-0.050	$[-0.075, -0.024]$	3×10^{-4}	1×10^{-3}

TABLE V

LORA RANK ABLATION (LORA + HEAD) AND THE BITFIT COMPARATOR, MEAN ± SD OVER SEEDS 42–46. SEQUENTIAL LINEAR SHOWN FOR REFERENCE.

Config	Dice↑	HD95↓	MAE (yr)	Trainable (T1)
BitFit	0.436 ± 0.031	77.1 ± 9.3	14.66	6,434
$r = 4$	0.754 ± 0.032	49.0 ± 6.6	14.32	23,502
$r = 8$	0.750 ± 0.027	48.8 ± 4.6	14.43	46,938
$r = 16$	$\mathbf{0.772 \pm 0.011}$	$\mathbf{44.8 \pm 2.4}$	14.67	93,810
Seq. Linear	0.782 ± 0.032	43.5 ± 7.1	14.71	8,446,306

A second cohort. OASIS-1 repeats both global tasks on a different scanner, a single T1 channel and a wider age range, and the pattern is identical: LoRA reaches brain-age MAE 11.95 ± 0.68 yr ($R^2 = 0.60$) against a 23.0 yr constant baseline, and sex AUC 0.920 ± 0.035; linear probing and BitFit return constants on both (MAE 23.0, balanced accuracy 0.50). Full fine-tuning is competitive here—it matches LoRA on brain age (Δ -0.05 yr, $p = 0.46$) and has slightly higher sex accuracy (-0.031, $p_{\text{Holm}} = 0.12$)—so the claim that survives both cohorts is the one about frozen strategies: with a small head, LoRA is the only frozen-backbone method that learns the task, and it is never worse than releasing all 14M parameters by more than the noise. Dementia classification (CDR>0, T9) is learned by none of the strategies at 32 labelled subjects (AUC 0.42–0.56, balanced accuracy 0.49–0.52), which bounds what the protocol can detect and is reported as such.

A second adapter family. Houlsby-style bottleneck adapters, the other standard way of moving a frozen encoder, behave like LoRA on the global tasks and unlike it on segmentation (Adapter column of Table VIII; paired contrasts in Tables III and IV). On the global tasks they learn everything LoRA learns: IXI brain age 11.48 ± 2.15 yr (Δ|err| $+0.49$ yr against LoRA, $p_{\text{Holm}} = 0.13$), OASIS brain age 9.68 ± 1.08 yr, which is 2.26 yr *better* than LoRA ($p = 1\times10^{-17}$), and sex AUC 0.903 and 0.915 (per-case accuracy 0.02 and 0.05 above LoRA, $p_{\text{Holm}} = 0.26$ and 1×10^{-3}). The small-head finding is therefore a statement about moving the encoder representation, not about LoRA's particular form. On segmentation the two diverge: adapters match LoRA on hippocampus (+0.002 Dice) and WMH (+0.003, $p_{\text{Holm}} = 1.0$) but lose 0.017 Dice on BraTS ($p = 5\times10^{-16}$) and 0.104 on stroke lesions ($p = 5\times10^{-61}$), the two tasks whose target is a small lesion. We note, without testing it, that a serial bottleneck rewrites every feature map it is inserted into whereas a parallel low-rank branch only adds to it; what the data establish is that the two PEFT families are interchangeable where the head is small and are not where a decoder must localise a small target.

TABLE VI

SHOT ABLATION, BRATS DICE, MEAN ± SD OVER SEEDS 42–46.

Method	$N_k = 16$	$N_k = 32$	$N_k = 64$
LoRA + head, r=16	0.646 ± 0.028	0.772 ± 0.011	0.816 ± 0.011
Sequential Linear	0.747 ± 0.030	0.782 ± 0.032	0.816 ± 0.018

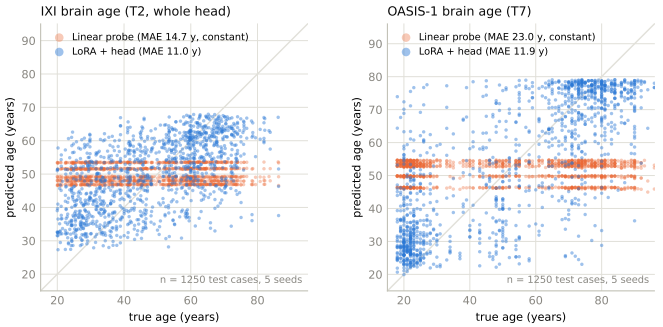


Fig. 3. Predicted against true brain age on the held-out test sets, all five seeds pooled (1250 cases per cohort), LoRA + head (r=16) against the linear probe on the same frozen encoder. Whole-head 96^3 input. The probe's output barely varies with the input (its spread is seed-to-seed, not case-to-case); LoRA's tracks age on IXI and OASIS-1 alike, with the compression towards the training mean expected at 32 labelled subjects.

F. How Far Below Full Data?

Table IX trains the same strategies on every labelled subject available outside the seed-42 validation and test sets, on the four tasks that have any. With 908 BraTS volumes LoRA + head reaches Dice 0.859, the linear probe 0.849 and full fine-tuning 0.865, against 0.772, 0.782 and 0.787 at 32 shots; with 249 IXI subjects LoRA's brain-age MAE falls from 10.99 to 8.33 yr (R^2 0.31 to 0.55) and its sex AUC from 0.878 to 0.994; with 106 OASIS subjects from 11.95 to 8.27 yr (R^2 0.60 to 0.79) and 0.920 to 0.971. Two things follow. The 32-shot results are not near saturation: they sit 0.09 Dice and 2.7–3.7 yr below what the same recipe reaches with all the data, which is the regime the paper is about. And the ordering between strategies survives the change of regime: on BraTS the probe is still within 0.01 Dice of LoRA and full fine-tuning still leads by less than 0.01, so 93.8K low-rank parameters stay within 0.006 Dice of training all 22.6M even at 908 volumes; on all four global tasks LoRA still matches or beats full fine-tuning (8.33 vs 8.50 yr, 8.27 vs 9.45, AUC 0.994 vs 0.958

TABLE VII
IXI BRAIN AGE WITH WHOLE-HEAD INPUT ($\mathbf{96^3}$) VERSUS $\mathbf{64^3}$ PATCHES, MEAN ± SD OVER SEEDS 42–46, 250 TEST SUBJECTS PER SEED. PRED. SD IS THE STANDARD DEVIATION OF THE PREDICTED AGES; THE TRUE SD IS 16.5 YR, SO A VALUE NEAR ZERO DENOTES A CONSTANT PREDICTOR.

	whole head (96^3)			patch
Method	MAE (yr)↓	R^2 ↑	pred. SD	MAE
LoRA + head, r=16	$\mathbf{10.99 \pm 1.27}$	$\mathbf{+0.31 \pm 0.15}$	10.56	14.67
Sequential FT	12.67 ± 0.69	$+0.13 \pm 0.05$	9.39	14.65
Sequential Linear	14.68 ± 0.22	-0.04 ± 0.04	0.13	14.71
BitFit	14.67 ± 0.24	-0.04 ± 0.04	0.11	14.66
constant predictor	14.39	0.00	0	14.39

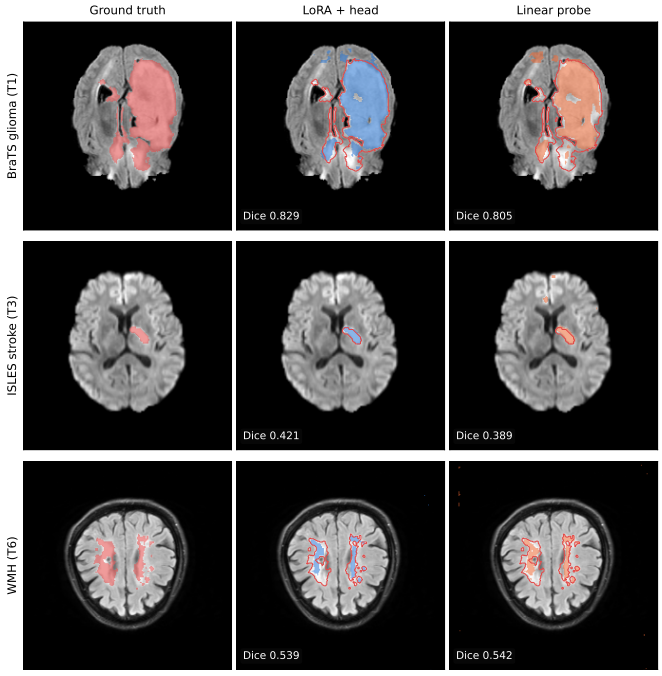


Fig. 4. Qualitative segmentation on held-out test cases (seed-42 split), one per task, chosen as the case whose LoRA Dice is closest to that task's median rather than its best. Left: ground truth (red). Middle and right: LoRA + head ($\boldsymbol{r}$=16) and the linear probe (decoder trained on the frozen encoder), with the ground-truth outline and the whole-volume per-case Dice. Slice through the largest ground-truth cross-section.

and 0.971 vs 0.953). These are single runs and are reported as reference points, not as a comparison.

G. Forgetting

Every method with a frozen backbone—adapter-isolated or linear-probed—has BWT exactly zero on every seed, because nothing shared is updated. Every method that updates the backbone forgets, and the mitigation baselines do not help: unregularised fine-tuning loses 0.331 ± 0.148 Dice on BraTS after learning brain age, EWC the same (-0.321 ± 0.181), and LwF (-0.475 ± 0.171) and replay (-0.400 ± 0.103) *more*. LwF is simultaneously the most accurate backbone-updating method on the first task (0.789) and the worst forgetter, which is the pattern to expect when the distillation term is computed on encoder features that the new task is free to move so long as the task loss outweighs it. At 32 shots, with tasks whose output types differ, none of these constraints is strong enough.

The three-task sequence sharpens this (Table X, Fig. 6). After learning ISLES, full fine-tuning's BraTS Dice falls from 0.779 ± 0.028 to 0.247 ± 0.280, and the variance is the point: the loss ranges from near zero to almost everything across seeds. A second segmentation task damages the first more than the heterogeneous regression task did, and how much is unpredictable. ISLES itself then loses 0.18 after brain age. Reversing the sequence (T2→T3→T1) shows the effect is symmetric: ISLES, now learned second, falls from 0.401 ± 0.085 to 0.010 ± 0.005 once BraTS is learned after it. Whichever segmentation task comes first, the next one erases it. The reversed two-task sequence (T2→T1) is uninformative on this point, because patch-based brain age sits at the constant baseline before anything can be forgotten (MAE 14.5 before, 14.6–14.8 after). The frozen methods score identically before and after every step, and ISLES Dice at learn is comparable across them (0.440 LoRA, 0.437 linear, 0.431 full fine-tuning).

H. What Each Strategy Costs to Deploy

Accuracy is one axis; a service that accumulates tasks also pays in storage, training time and hardware. Table XI gives the measured figures for the segmentation configuration, which is the largest of our tasks.

Storage is where the strategies separate. A LoRA adapter with its output head occupies 366 KB on disk, so a site running five tasks stores one 87 MB backbone plus 1.8 MB of adapters (89 MB in all). Linear probing must keep a separate 32 MB decoder per task (161 MB of decoders for five, 248 MB with the backbone), and full fine-tuning a separate 86 MB network per task (431 MB), since nothing is shared once every weight has moved. The ratio grows linearly with the number of tasks, which is the regime a long-lived service is in.

Training time runs the other way, and we report it plainly because it is counter-intuitive. The two strategies that update the fewest parameters are the *slowest* to train: BitFit 48.1 min and LoRA 41.5 min, against 15.8 for linear probing and 15.9 for full fine-tuning. Gradients must still be propagated through the entire network to reach parameters scattered across it, and LoRA's parallel low-rank branches add forward cost; only the optimiser state shrinks. Where linear probing back-propagates no further than the decoder, LoRA and BitFit must traverse the encoder as well. Parameter efficiency here buys storage and task isolation, not training speed—a distinction often elided when PEFT is described as "cheap". All four fit on one consumer GPU, peaking between 0.9 and 1.7 GB, so hardware is not the discriminator at this scale.

I. Does the Pretrained Checkpoint Help?

We repeated the LoRA + head (r=16) configuration on every task with the backbone randomly initialised instead of loaded from the FOMO checkpoint, changing nothing else (Fig. 7; RI column of Table VIII; paired contrasts in Tables III and IV). The answer is task-dependent and, where present, small. On glioma segmentation the two are identical ($+0.000$ Dice, 95% CI $[-0.002, +0.002]$, $p = 0.23$), and on IXI brain

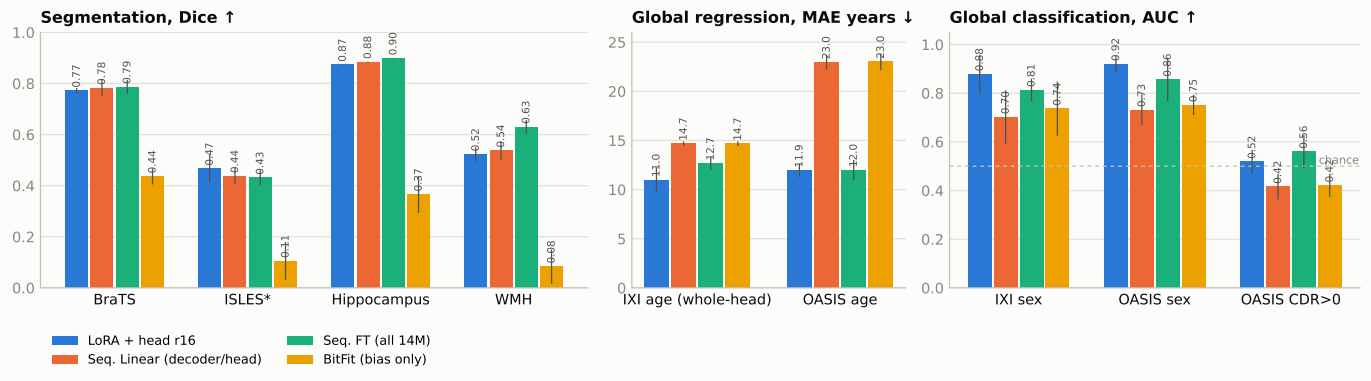


Fig. 5. Which strategy wins depends on whether the task inherits a large task head. Four strategies on all nine tasks at $N_k = 32$, mean ± SD over five seeds. Left: segmentation tasks (Dice; ISLES Sequential Linear and FT are scored at learn inside the three-task sequence). Middle and right: global tasks (MAE in years; AUC), where linear probing and BitFit return constant outputs. Dashed line: chance.

TABLE VIII

ALL NINE TASKS, $N_k = 32$, MEAN ± SD OVER SEEDS 42–46. SEGMENTATION REPORTS DICE↑ ; BRAIN AGE MAE IN YEARS↓; CLASSIFICATION AUC↑ . "CONST." MARKS A METHOD WHOSE OUTPUT DOES NOT VARY WITH THE INPUT (PREDICTION SD < 0.5 YR FOR BRAIN AGE; BALANCED ACCURACY WITHIN 0.02 OF 0.500 FOR CLASSIFICATION). RI = LORA + HEAD ON A RANDOMLY INITIALISED BACKBONE (SEC. V-I). ISLES SEQUENTIAL LINEAR AND FT ARE SCORED AT LEARN INSIDE THE THREE-TASK SEQUENCE. ADAPTER = HOULSBY-STYLE BOTTLENECK ADAPTERS AT LORA'S BUDGET (SEC. IV-D). BOLD: BEST OF THE FIVE STRATEGIES.

Task	Type	LoRA + head, r=16	Sequential Linear	Sequential FT	BitFit	Adapter	RI
Tasks with a large task-specific decoder (Dice↑)							
T1 BraTS glioma	seg, 3 ch	0.772 ± 0.011	0.782 ± 0.032	$\mathbf{0.787 \pm 0.028}$	0.436 ± 0.031	0.755 ± 0.046	0.772 ± 0.029
T3 ISLES stroke	seg, 3 ch	$\mathbf{0.440 \pm 0.033}$	0.437 ± 0.033	0.431 ± 0.029	0.105 ± 0.074	0.363 ± 0.048	0.496 ± 0.022
T4 MSD hippocampus	seg, 1 ch	0.873 ± 0.002	0.884 ± 0.002	$\mathbf{0.898 \pm 0.001}$	0.365 ± 0.071	0.875 ± 0.002	0.871 ± 0.002
T6 WMH 2017	seg, 2 ch, 5 scanners	0.522 ± 0.027	0.537 ± 0.036	$\mathbf{0.630 \pm 0.027}$	0.083 ± 0.067	0.525 ± 0.037	0.499 ± 0.067
Tasks with only a ∼1K-parameter head							
T2 IXI brain age	reg, MAE yr↓	$\mathbf{10.99 \pm 1.27}$	14.68 (const.)	12.67 ± 0.69	14.67 (const.)	11.48 ± 2.15	10.99 ± 0.75
T7 OASIS brain age	reg, MAE yr↓	11.95 ± 0.68	22.96 (const.)	12.00 ± 1.01	23.01 (const.)	$\mathbf{9.68 \pm 1.08}$	13.10 ± 1.22
T5 IXI sex	cls, AUC↑	0.878 ± 0.079	0.702 (const.)	0.814 ± 0.047	0.737 (const.)	$\mathbf{0.903 \pm 0.070}$	0.847 ± 0.061
T8 OASIS sex	cls, AUC↑	$\mathbf{0.920 \pm 0.035}$	0.731 (const.)	0.856 ± 0.090	0.752 (const.)	0.915 ± 0.030	0.879 ± 0.048
T9 OASIS CDR>0	cls, AUC↑	0.518 (const.)	0.416 (const.)	0.560 (const.)	0.421 (const.)	0.456 (const.)	0.533 (const.)
Trainable (seg / global)		93.8K / 56.4K	8.45M / 1.0K	22.6M / 14.1M	6.4K / 5.0K	97.5K / 66.1K	93.8K / 56.4K

TABLE IX

FULL-DATA REFERENCE, SEED 42: THE 32-SHOT RESULT (MEAN ± SD OVER FIVE SEEDS) BESIDE THE SAME STRATEGY TRAINED ON EVERY LABELLED SUBJECT OUTSIDE THAT SEED'S VALIDATION AND TEST SETS (N_{FULL}; ONE RUN). BRATS RUNS 10 EPOCHS OVER 908 VOLUMES; THE GLOBAL TASKS KEEP THE 100-EPOCH RECIPE. SEQUENTIAL LINEAR IS CONSTANT ON THE GLOBAL TASKS AT 32 SHOTS AND WAS NOT RERUN. ISLES, HIPPOCAMPUS, WMH AND CDR HAVE NO SUBJECTS OUTSIDE THE FEW-SHOT PROTOCOL'S SPLITS.

			LoRA + head, r=16		Sequential Linear		Sequential FT	
Task	Metric	N_{full}	32 shots	full	32 shots	full	32 shots	full
T1 BraTS glioma	Dice↑	908	0.772 ± 0.011	0.859	0.782 ± 0.032	0.849	0.787 ± 0.028	0.865
T2 IXI brain age	MAE yr↓	249	10.99 ± 1.27	8.33	14.68 ± 0.22	—	12.67 ± 0.69	8.50
T5 IXI sex	AUC↑	249	0.878 ± 0.079	0.994	0.702 ± 0.110	—	0.814 ± 0.047	0.958
T7 OASIS brain age	MAE yr↓	106	11.95 ± 0.68	8.27	22.96 ± 0.75	—	12.00 ± 1.01	9.45
T8 OASIS sex	AUC↑	106	0.920 ± 0.035	0.971	0.731 ± 0.062	—	0.856 ± 0.090	0.953

age likewise (-0.00 yr, $p = 0.51$). On stroke lesions random initialisation is *better*, by 0.030 Dice ($p = 1 \times 10^{-12}$). Pretraining helps on hippocampus (+0.002 Dice), WMH (+0.024), OASIS brain age (1.2 yr) and both sex tasks (+0.11 and +0.03 accuracy), all at $p_{\text{Holm}} \leq 0.013$.

We report this plainly: with this checkpoint, at this shot count, transfer is real on some tasks and absent on others, and its largest effect (0.11 accuracy on IXI sex) is of the same order as the differences between adaptation strategies, while on the two original tasks it is zero. What the rest of the paper demonstrates is therefore parameter-efficient adaptation of a frozen network: the decision rule between strategies does not depend on what the frozen weights contain, since it reproduces on a randomly initialised backbone, and our results should not be read as a measurement of the FOMO corpus's value. Whether a different checkpoint, a larger backbone, or unfreezing during adaptation would change this is open; what is settled is that the transfer cannot be assumed, and that a randomly initialised control is cheap enough to run every time.

TABLE X

THREE-TASK SEQUENCES AT $N_k = 32$. TOP: T1→T3→T2 (BRATS → ISLES → IXI BRAIN AGE, 64^3 PATCHES), MEAN ± SD OVER SEEDS 42–46. BOTTOM: THE REVERSED ORDER T2→T3→T1, SEEDS 42–44. EACH TASK IS SCORED ON ITS OWN HELD-OUT TEST SET WHEN LEARNED AND AGAIN AFTER EACH LATER TASK. ISLES TEST SET: 155 SUBJECTS.

Method	At learn			BraTS Dice after…		ISLES Dice
	BraTS Dice	ISLES Dice	Age MAE	ISLES	age	after age
LoRA + head, r=16	0.773 ± 0.011	0.440 ± 0.033	14.51 ± 0.48	=	=	=
Sequential Linear	0.782 ± 0.032	0.437 ± 0.033	14.75 ± 0.26	0.782 ± 0.032	0.782 ± 0.032	0.437 ± 0.033
Sequential FT	0.779 ± 0.028	0.431 ± 0.029	14.69 ± 0.22	0.202 ± 0.172	0.247 ± 0.280	0.254 ± 0.073
Method	At learn (reversed)			Age MAE after…		ISLES Dice
	Age MAE	ISLES Dice	BraTS Dice	ISLES	BraTS	after BraTS
LoRA + head, r=16	14.57 ± 0.44	0.478 ± 0.031	0.793 ± 0.020	=	=	=
Sequential FT	14.56 ± 0.09	0.401 ± 0.085	0.779 ± 0.049	14.39 ± 0.08	14.42 ± 0.18	0.010 ± 0.005

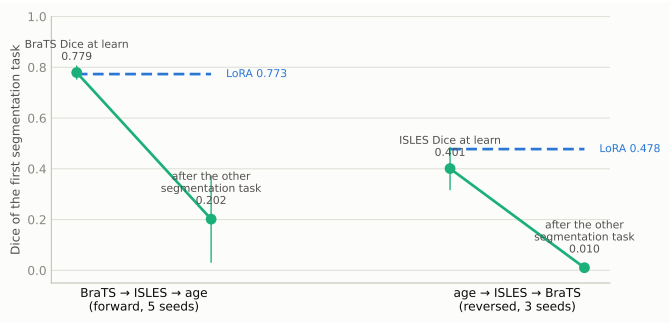


Fig. 6. Under full fine-tuning, the first segmentation task in a three-task sequence is erased by the second whichever order the tasks arrive in: BraTS Dice $\mathbf{0.78}\rightarrow\mathbf{0.20}$ after ISLES and brain age (forward, five seeds); ISLES Dice $\mathbf{0.40}\rightarrow\mathbf{0.01}$ after BraTS (reversed, three seeds). Dashed: LoRA + head on the same task, which by construction is unchanged by later tasks.

TABLE XI

DEPLOYMENT COST, MEASURED ON THE SEGMENTATION CONFIGURATION. STORAGE IS EXACT; TRAINING TIME IS THE MEAN OVER SEEDS 42–44, WHICH RAN ON AN OTHERWISE IDLE GPU (SEEDS 45–46 SHARED THE CARD WITH OTHER JOBS AND ARE EXCLUDED FROM TIMING; THEIR ORDERING IS THE SAME). "PER TASK" IS WHAT MUST BE STORED FOR EACH ADDITIONAL TASK ONCE THE SHARED BACKBONE (87 MB) IS ON DISK; FULL FINE-TUNING SHARES NOTHING, SO ITS BACKBONE IS COUNTED PER TASK.

Strategy	Trainable	Per task	5 tasks	Train
LoRA + head, r=16	93.8 K	**366 KB**	**89 MB**	41.5 min
Linear probe (decoder)	8.45 M	32.2 MB	248 MB	**15.8 min**
Full fine-tuning	22.6 M	86.2 MB	431 MB	15.9 min
BitFit	6.4 K	25 KB	87 MB	48.1 min

TABLE XII

SELECTING A STRATEGY FOR A NEW TASK. THE CONDITION IS KNOWABLE BEFORE ANY TRAINING: IT ASKS WHETHER THE ARCHITECTURE ALREADY CONTAINS A LARGE MODULE SHAPED FOR THE NEW TASK'S OUTPUT.

Situation	Recommendation
New task reuses a large existing head (e.g. a segmentation decoder)	Train that head on a frozen encoder. It is the most accurate frozen option and the simplest. Prefer LoRA only if per-task storage matters: it costs at most 0.015 Dice (none on two of four tasks) and saves 88× the disk. If the task is the only one the model will ever serve, full fine-tuning is more accurate still. Bottleneck adapters are not a substitute for LoRA here: they lose up to 0.10 Dice where the target is a small lesion.
New task has only a small head (a pooled regressor or classifier)	Move the encoder representation: LoRA or bottleneck adapters, which perform alike here. A small head on frozen features returns a constant output even when those features carry usable signal; this is the one case where adapters are necessary rather than economical.
Prior tasks must not degrade	Freeze the shared weights, by either route. Do not rely on EWC, LwF or replay: EWC forgot as much as unregularised fine-tuning here and LwF and replay forgot more, and a second segmentation task erased the first in either order.
More than ∼64 annotated cases available	The accuracy gap between frozen strategies closes; choose on storage and operational fit rather than on Dice.
Adopting a pretrained checkpoint	Validate the transfer with a random-initialisation control before assuming it. Ours made no difference on two tasks, hurt on one and helped modestly on six (Sec. V-I).

VI. DISCUSSION

A. Choosing a Strategy

Table XII states the recommendations compactly; this section gives the reasoning behind them.

Our two task families answer this in opposite directions, each replicated on a task chosen to share little with the original, and the contrast is the useful part.

When the task head is large, adapters buy economy. Segmentation inherits an 8.45M-parameter decoder that is already task-shaped; training it is a strong and simple baseline, and LoRA + head stays within 0.015 Dice of it on four tasks that otherwise share almost nothing—three modalities, two or one; pathology or healthy anatomy; a lesion occupying <0.1% of voxels or a structure occupying 4.7%; one scanner or five—while training 90× fewer parameters, and closing to equality by 64 shots. The stability of that margin across such different tasks is itself the evidence that what matters is the decoder, not the imaging problem. Here the argument for adapters is cost—93.8K parameters and under 400 KB per task on disk, against a 32 MB decoder per task—not accuracy; and where a model will only ever serve one task, full fine-tuning is more accurate than either (by 0.11 Dice on WMH). The margin also survives more data: at 908 BraTS volumes LoRA, the probe and full fine-tuning are within 0.016 Dice of one another. Not every adapter family qualifies, though: bottleneck adapters at

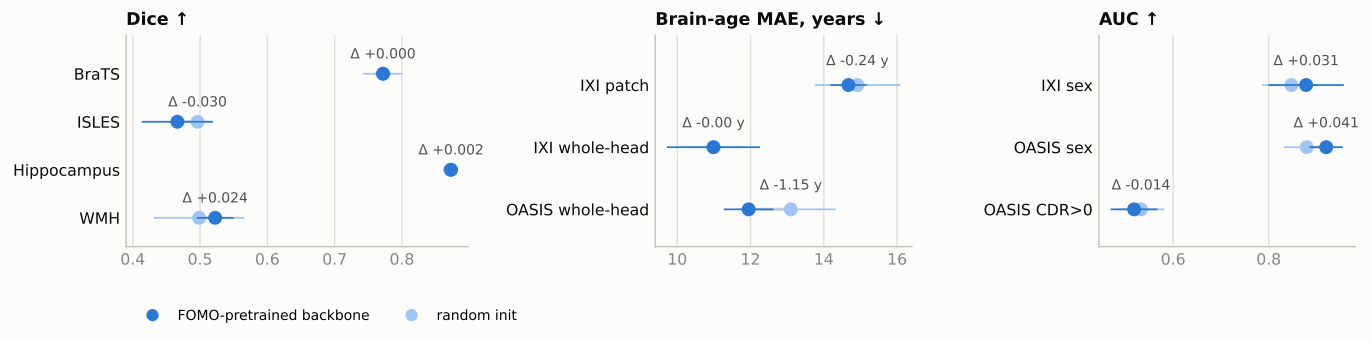


Fig. 7. Pretrained (FOMO checkpoint) versus randomly initialised backbone under the same LoRA + head ($r=16$) configuration on all nine tasks, mean ± SD over five seeds; **Δ** is pretrained minus random. Paired per-case tests are in Tables III and IV.

the same budget match LoRA where the foreground is large (hippocampus, WMH) and lose 0.017 and 0.104 Dice on the two small-lesion tasks.

When there is no task head to speak of, adapters are necessary. Brain age and sex both attach a ∼1K-parameter head to pooled encoder features. On whole-head input neither is learnable from frozen features, on either cohort, and bias-only adaptation does not help: both emit a constant—a fixed age, or a single class on every subject. LoRA's 56K encoder parameters learn both on both cohorts, better than releasing all 14M on IXI and as well as it on OASIS. Bottleneck adapters with 66K do the same, and on OASIS brain age better still (9.7 yr), which places the finding with encoder adaptation as such rather than with LoRA.

Sex classification says something the regression result could not. Linear probing's AUC is 0.70–0.73 and BitFit's 0.74–0.75 on the two cohorts: the frozen features rank subjects by sex far better than chance, so the information is *present*. What the 1K-parameter head cannot do is convert that ranking into a decision—its balanced accuracy is 0.500. The bottleneck is therefore not missing information but an unusable geometry, and moving the representation is what makes the signal exploitable. This is the clearest statement we can give of why parameter-efficient adaptation is not interchangeable with a bigger readout.

No replay means no retained prior-task data. Every strategy we recommend trains on the new task's cases alone. Replay-based continual learning requires keeping examples from earlier tasks available for the lifetime of the service, which under HIPAA or GDPR is a governance commitment rather than a storage one; in our results it also did not work, forgetting more than doing nothing. Freezing obtains the same protection with no retention at all.

Forgetting-free is a property of freezing, not of adapters. Adapter isolation and frozen-encoder probing both have BWT zero; the distinction between them is Table I, not the BWT column. What is worth reporting is the other side: none of EWC, LwF or replay achieved it—EWC matched plain fine-tuning and LwF and replay were worse at this scale—and a second task of the *same* type as the first erased it whichever came first, with full fine-tuning's loss ranging from near zero to almost all of the Dice across seeds. If prior-task performance matters, freezing is the reliable route and the standard remedies are not.

The 66-parameter lesson. Whether the task's output projection is trainable moves Dice by 0.22. This is not a subtle finding, but it is an easy configuration error: it is what a literal reading of "freeze the backbone, train the adapter" produces, and on its own it is enough to make LoRA look unsuited to segmentation.

B. Limitations

Pretraining helps inconsistently, and this generalises only as a method. The control (Sec. V-I) finds the checkpoint indistinguishable from random initialisation on the two original tasks, worse on one and modestly better on six. Our results therefore support parameter-efficient adaptation of a frozen backbone, and the strategy comparison reproduces on a random backbone; they do not measure the value of large-scale pretraining, and the reader should not generalise the specific per-task findings to other checkpoints or scales. What does generalise is the practice: a randomly initialised control costs one additional training run and is the only way to know whether a pretrained model is contributing anything to a given deployment.

Nine tasks, but six data sources and one imaging domain. Brain age and sex share subjects and scans within each cohort, so within a cohort they are evidence about output type, not about data; T2 and T7 (and T5 and T8) are the independent replications. All nine tasks are brain MRI, and all share one backbone. Whether the pattern holds for other anatomy, modalities or architectures is untested.

Five seeds, one architecture, one shot count for most results. Means and SDs are over five training runs (three for the reversed sequences); the paired tests pool cases across those runs and test the per-case difference, not run-to-run variability. Only the shot ablation spans 16–64 shots, and only on two methods.

Task identity is given at inference. Adapter selection assumes the task is known; task-agnostic routing is not addressed, and it is the obvious next step for a deployed system.

Two sequences, two orders each. T1→T2 and T1→T3→T2 and their reversals are the only orders evaluated; a sequence longer than three tasks, or one containing two global tasks, is not.

Brain age is improved, not solved. MAE 11.0 yr on IXI ($R^2 = 0.31$) and 12.0 yr on OASIS ($R^2 = 0.60$) beat a constant by a wide margin but are far from the 3–4 yr that large supervised cohorts achieve; 32 labeled subjects and a 96^3 resampling are the obvious constraints. Dementia from 32 labelled T1 volumes (T9) is not learnable by any strategy here, which marks the floor of the protocol rather than a property of any method.

The full-data references are single runs on four tasks. They bound the 32-shot results from above and show that the ordering between strategies survives more data, but with one seed each they cannot support paired tests, and five of the nine tasks have no subjects outside the few-shot protocol's splits, so their ceilings are unknown.

Most tasks are single-task evaluations. T4–T9 are run on their own rather than inside a continual sequence; they test the accuracy patterns, not forgetting, which the three-task sequences cover.

Infarct detection as classification excluded. A binary infarct-detection formulation contrasting ISLES with IXI controls was prepared but is not reported, and the reason is worth stating because the design is a natural one: the positive class carries DWI, FLAIR and ADC channels and the negative class T1 and T2 with zero-filled channels, so the classes differ by acquisition rather than pathology and any classifier score would be uninterpretable. We use ISLES for lesion segmentation instead, where the cohort is internally consistent.

No augmentation. Few-shot results would likely improve with it for all methods; we omit it to keep the method comparison clean.

VII. Conclusion

We compared ten adaptation strategies across nine brain-MRI tasks, six cohorts, three output types and five seeds, under a protocol with subject-grouped held-out splits and whole-volume inference, to answer a question a deployed imaging service actually faces: how should a new task be added when only a few dozen annotations exist and prior-task data is unavailable?

The choice is conditional on a property of the new task that can be checked before any training. Tasks that reuse a large existing head—all four segmentation tasks here—are well served by training that head on a frozen encoder; LoRA trails it by at most 0.015 Dice and matches it on two of the four, across tasks differing in modality count, pathology, scale and scanner, and is preferable only when per-task storage matters, where it saves 88× the disk. Tasks with only a small head—brain age and sex, from whole-head input, on two cohorts—cannot be learned that way at all: linear probing and bias-only tuning return a constant output even though their frozen features rank subjects well above chance, and LoRA is the sole frozen strategy that works, with 56 K parameters matching or beating full fine-tuning of 14 M. Where prior-task performance must be preserved, freezing is the only route that delivered it: a second segmentation task erased the first under full fine-tuning in either order, EWC did not help, and LwF and replay made it worse.

Three cautions accompany these recommendations. Parameter efficiency buys storage and isolation, not training speed—the two methods with the fewest trainable parameters were the slowest to train. A pretrained checkpoint should be validated with a random-initialisation control rather than assumed to transfer; ours made no difference on the two tasks we started with and a small one elsewhere. And an evaluation protocol that scores random crops on the model-selection split can inflate results enough to reverse the ordering of methods, which is why we release ours along with the code, the splits and the per-case predictions.

Code and Data Availability

The code, the persisted subject-grouped splits for every task and seed, the per-case predictions behind every table and figure, the table and statistics generators, and the protocol tests are available at `https://github.com/ChiShengChen/FewShot3DBrain`. The imaging data are obtained from their original sources under the terms listed in Sec. IV-A; no imaging data are redistributed.

[1] F. C. Organizers, "FOMO-60K: Foundation models for open medical imaging," 2025, fOMO25 Challenge, HuggingFace: FOMO25/FOMO-MRI.

[2] FOMO25, "Fomo baseline codebase," 2025. [Online]. Available: https://github.com/fomo25/baseline-codebase

[3] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska *et al.*, "Overcoming catastrophic forgetting in neural networks," *PNAS*, vol. 114, no. 13, pp. 3521–3526, 2017.

[4] M. McCloskey and N. J. Cohen, "Catastrophic forgetting in connectionist networks: The sequential learning problem," *Psychology of learning and motivation*, vol. 24, pp. 109–165, 1989.

[5] Z. Li and D. Hoiem, "Learning without forgetting," in *ECCV*, 2016.

[6] E. B. Zaken, S. Ravfogel, and Y. Goldberg, "BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models," in *ACL*, 2022.

[7] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *ICLR*, 2022.

[8] J. Y.-C. Hu, M. Su, E.-J. Kuo, Z. Song, and H. Liu, "Computational limits of low-rank adaptation (lora) fine-tuning for transformer models," *arXiv preprint arXiv:2406.03136*, 2024.

[9] A. Hatamizadeh *et al.*, "Unetr: Transformers for 3d medical image segmentation," *WACV*, 2022.

[10] H. Wang *et al.*, "SAM-Med3D," *arXiv preprint arXiv:2310.15161*, 2023.

[11] R. J. Chen *et al.*, "Towards a visual-language foundation model for computational pathology," *Nature Medicine*, vol. 30, pp. 863–874, 2024.

[12] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," *ICML*, 2017.

[13] S.-A. Rebuffi, N. Komodakis *et al.*, "iCARL: Incremental classifier and representation learning," in *CVPR*, 2017.

[14] M. M. Derakhshani *et al.*, "Lifelonger: A benchmark for continual disease classification," *MICCAI*, 2022.

[15] C. González *et al.*, "Lifelong nnU-Net: A framework for standardized medical continual learning," *Scientific Reports*, vol. 13, 2023.

[16] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, "Parameter-efficient transfer learning for nlp," *ICML*, 2019.

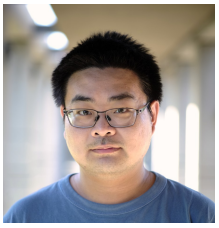
[17] X. L. Li and P. Liang, "Prefix-tuning: Optimizing continuous prompts for generation," in *ACL*, 2021.

[18] A. Mallya and S. Lazebnik, "PackNet: Adding multiple tasks to a single network by iterative pruning," *CVPR*, 2018.

[19] A. Mallya, D. Davis, and S. Lazebnik, "Piggyback: Adapting a single network to multiple tasks by learning to mask weights," *ECCV*, 2018.

[20] H. Face, "Peft: State-of-the-art parameter-efficient fine-tuning methods," 2023. [Online]. Available: https://github.com/huggingface/peft

[21] B. Menze *et al.*, "The BraTS 2023 challenge," in *MICCAI BraTS*, 2023.
[22] I. C. London and G. Hospital, "IXI dataset," 2008, brain Development, research use. [Online]. Available: https://brain-development.org/ixi-dataset/
[23] M. R. Hernandez Petzsche, E. de la Rosa, U. Hanning *et al.*, "ISLES 2022: A multi-center magnetic resonance imaging stroke lesion segmentation dataset," *Scientific Data*, vol. 9, no. 1, p. 762, 2022.
[24] M. Antonelli, A. Reinke, S. Bakas *et al.*, "The medical segmentation decathlon," *Nature Communications*, vol. 13, no. 1, p. 4128, 2022.
[25] H. J. Kuijf, J. M. Biesbroek, J. de Bresser, R. Heinen, S. Andermatt, M. Bento, M. Berseth, M. Belyaev, M. J. Cardoso, A. Casamitjana *et al.*, "Standardized assessment of automatic segmentation of white matter hyperintensities and results of the WMH segmentation challenge," *IEEE Transactions on Medical Imaging*, vol. 38, no. 11, pp. 2556–2568, 2019.
[26] D. S. Marcus, T. H. Wang, J. Parker, J. G. Csernansky, J. C. Morris, and R. L. Buckner, "Open access series of imaging studies (OASIS): Cross-sectional MRI data in young, middle aged, nondemented, and demented older adults," *Journal of Cognitive Neuroscience*, vol. 19, no. 9, pp. 1498–1507, 2007.



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